1 Verification

Each v0.1 report in the table can be checked with one command, from an empty directory,

```
uvx agent-evidence-vectors==0.32.1 -read-w3c-report <file>
```

which prints `conforms`, or one line per shape error or rejection row, and exits 0 when the report conforms, 1 when it does not, and 2 when the reader reached no verdict. Rows that are not reports (readers' verdicts, comparisons, census counts) are checked by their rerun command instead. The evidence classes are their authors' words: "reproduction in O8's sense" (Oct/0163), "a reader written from the 0087 text alone" and "proposed" (Oct/0162), "the closest this paper comes to an independent record" (Oct/0150). The class of every cell this appendix runs is "automated rerun from the pins, operated by the appendix's author". Following Oct/0163, units differ by row and no column is summed.

Table 1: The verification column. Commands run in the appendix directory unless they start with `uvx`.

Row	Command	Verdict	Unit	Denominator	Evidence class
R1	`-read-w3c-report report.json`	conforms	checks	272	reproduction in O8's sense (Oct/0170, Oct/0171)
R2	`-read-w3c-report` on each of the three reports	conforms	checks	72, 168, 750	reproduction in O8's sense (Oct/0064)
R3-v01	`-read-w3c-report run-report-v01.json`	conforms	checks	48	producer's converter (Nov/0000)
R4	`-read-w3c-report` on each of the three reports	conforms from v0.18.1; rejected under W3C-R-020 before v0.18	checks	76, 67, 16	the closest to an independent record, through the producer's glue (Oct/0150)
R4-driver	`python rerun.py R4-driver`	equal to the recorded run	cases	as recorded	without the glue (Nov/0003)
R5	`-read-w3c-report finality-threshold-report.level1.json`	conforms	draws against thresholds	150,006	reproduction in O8's sense (Oct/0083, Oct/0084)
R6	`python rerun.py R6`	reproduced; no report file	tests	4 baseline, 6 claim-boundary	reproduction in O8's sense (Oct/0018)
R6, re-encoded	`-read-w3c-report run-report-v01.json`	conforms	checks	4	producer's converter (Nov/0006)
R7	`python rerun.py R7`	reproduced; a native packet, not a v0.1 report	outcomes	21	reproduction in O8's sense (Oct/0085, Oct/0184)
R8	`python rerun.py R8`	reproduced	members	169 judged	a reader written from the 0087 text alone
R8-compare	`python rerun.py R8-compare`	92 of 96; 23, 2, 44 of 69	members	96 and 69	a comparison (Oct/0188, Nov/0002)
R9	`python rerun.py R9`	83/82 at v0.17.4; 41/124 at v0.18.0 and v0.18.1	members	165	a comparison across releases
R10	`python rerun.py R10` (added)	bytes as posted	values	1	posted measurement (Oct/0038, Oct/0046)
R11	`python rerun.py R11`	output digest as posted	members	165	a reader written from the text
R12	`python rerun.py R12`	per-member verdicts as posted	members x paths	33 x 3	proposed
R13	`python rerun.py R13`	every listed file OK	synthetic cases	5	reproduction in O8's sense (Oct/0179); proposed
R14	`python matrix/matrix.py run`	see the matrix below	verdicts	reports x readers	a comparison
R15	none posted	tally only: 3 of 592	artifacts	592	the producer's own count (Oct/0162)

Every report under every reader. The reader matrix reruns R1, R2, R3, R3-v01, R4, R5 and R6 from their pins, keeps the nine files those reruns write that declare the v0.1 format and the report R6's producer posted beside its commands, ten in all, and reads each with eleven readers: the Python reference reader at 0.12.0, 0.13.0, 0.17.4, 0.18.1 and 0.31.0; the Go and Rust readers at v0.18.1 and at the current commit; and the two readers written from

Reader	R1	R2-v0.2	R2-v0.3	R2-v0.4	R3-v01	R4-heldfast	R4-native	R4-map	R5	R6	R6-v01
Python 0.12.0	C	C	C	C	C	R R-020	R R-020	R R-020	C	R shape	C
Python 0.13.0	C	C	C	C	C	R R-020	R R-020	R R-020	C	R shape	C
Python 0.17.4	C	C	C	C	C	R R-020	R R-020	R R-020	C	R shape	C
Python 0.18.1	C	C	C	C	C	C	C	C	C	R shape	C
Python 0.31.0	C	C	C	C	C	C	C	C	C	R shape	C
Go v0.18.1	C	C	C	C	C	C	C	C	C	R shape	C
Go main	C	C	C	C	C	C	C	C	C	R shape	C
Rust v0.18.1	C	C	C	C	C	C	C	C	C	R shape	C
Rust main	C	C	C	C	C	C	C	C	C	R shape	C
0039 reader	C	C	C	C	C	C	C	C	C	N	C
Kroehl 6d8ee54b	C	C	C	C	C	C	C	C	C	T	C

Table 2: Each report rerun from its pins, read by each reader. C conforms; R rejected, with the rows cited (a numbered row of the rejection table by its number, an unnumbered rule as R- and its identifier's number); T cannot tell; N cannot read. R7 has no column: its packet is not a v0.1 report.

Reader	Mutants	Rejected	Conforms	Cannot tell	Cannot read	Target cited
Python 0.12.0	168	161	7	0	0	160/168
Python 0.13.0	168	161	7	0	0	161/168
Python 0.17.4	168	161	7	0	0	161/168
Python 0.18.1	168	168	0	0	0	168/168
Python 0.31.0	168	168	0	0	0	168/168
Go v0.18.1	168	168	0	0	0	168/168
Go main	168	168	0	0	0	168/168
Rust v0.18.1	168	168	0	0	0	159/168
Rust main	168	168	0	0	0	159/168
0039 reader	168	61	88	18	1	58/59
Kroehl 6d8ee54b	168	82	37	49	0	58/59

Table 3: Single-field negative controls: every reader on every mutant. Target cited counts mutants whose target row the reader named, over those whose target its labels can name.

the text, Rouiada Abbas's of Oct/0039 and Lars Kersten Kroehl's at 6d8ee54b. Those two read corpus members, so each report is wrapped as a member with an adapter that writes `{"id": <input>, "subject": <report>}` and nothing else. Every report conforms under every reader except R4's three, which the Python readers of 0.12.0 to 0.17.4 reject under W3C-R-020, and R6's, which names the format in prose and carries records in place of checks: every Python, Go and Rust reader rejects it on shape, Rouiada Abbas's reader cannot read it and Lars Kersten Kroehl's cannot tell. No cell contradicts a verdict posted on the list (R14). Roel Schuurkes's re-encoded R6 (Nov/0006, sha256 `699a52b3...`) came after this run and is not in the table yet; the Python reader at 0.32.1 prints conforms on it, as v0.18.1 does in his run.

- R1: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report report.json`
- R2-v0.2: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-residue-v0.2.json`
- R2-v0.3: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-residue-v0.3.json`
- R2-v0.4: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-residue-v0.4.json`
- R3-v01: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-report-v01.json`
- R4-heldfast: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report heldfast-profiles.json`
- R4-native: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report native-verify-mjs.json`
- R4-map: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report probity-map-reader.json`
- R5: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report finality-threshold-report.level1.json`
- R6: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report records.v0.1.json`
- R6-v01: `uvx agent-evidence-vectors==0.32.1 -read-w3c-report run-report-v01.json`

Negative controls for the readers. A reader that accepted everything would pass R14, so each reader also reads single-field copies of the reports, each aimed at one rejection row: 154 copies per reader, plus 7 positive controls whose check identity is non-ASCII and whose root is recomputed over the RFC 8785 form. The Python reader from 0.18.1 on and the Go and Rust readers reject 154 of 154 and accept the 7 positive controls. The Python readers of 0.12.0 to 0.17.4 reject 147; they accept the 7 copies whose root is computed over the ASCII-escaped form and reject the 7 positive controls, the cause Kenne Ives gave for R4 (Oct/0150). A declared moved that disagrees with the recomputed value is aimed at W3C-R-021; a recomputed moved outside compared is W3C-R-016.

The readers written from the text answer cannot-tell where the text leaves the reading open, as the reader of 0039 was built to (Oct/0039), so their results are split by what the copy is aimed at. These are findings for their authors, not faults, and go to them first.

Table 4: The two readers written from the text, on the 154 copies and 7 positive controls, by target: rejected / conforms / cannot tell.

Copies aimed at	Rouiada Abbas's reader (Oct/0039)	Lars Kersten Kroehl's reader (6d8ee54b)
Rows 1 to 14 (56 copies)	55 / 1 / 0	55 / 0 / 1
Unnumbered requirements (81 copies)	3 / 70 / 8	17 / 26 / 38
Not v0.1-shaped (17 copies)	0 / 8 / 8, 1 cannot read	3 / 8 / 6
Positive controls (7)	0 / 7 / 0	0 / 7 / 0

On rows 1 to 14 the one copy neither reader rejects is the same: R1 with a qualifier on a non-verdict check, aimed at row 6.